

Nifty 100 Financial Intelligence

Analyst Guide — D-22

Purpose	Practical guide for analysts using the project's financial intelligence outputs and API.
Coverage	93 companies in the current unified financial-ratio database; 11 sectors.
Primary database	database/financial_data.db
API	FastAPI server on localhost:8000
Dashboard	Streamlit application on localhost:8501
Units	Monetary values are stored in INR Crore.

1. What this project provides

- Company-level financial ratios and latest financial information.
- Profitability, leverage, growth, cash-flow and capital-allocation intelligence.
- Company clustering into financial-profile groups.
- Screening based on analyst-selected financial thresholds.
- Sector and peer-company exploration.
- A FastAPI interface for programmatic access to the same intelligence.

2. Getting started

Open PowerShell in the project root and make sure the database exists. The current project uses **database/financial_data.db**. The API can then be started with the project's API command.

```
make api
```

The API is expected at **http://127.0.0.1:8000**. The Streamlit dashboard is launched with:

```
make dashboard
```

The dashboard is expected at **http://localhost:8501**.

3. API analyst workflow

Endpoint	Use
/health	Confirm that the API is running and the database is reachable.
/summary	Get project-level counts: companies, financial-ratio rows and sectors.
/companies	List the available company IDs.
/companies/{company}/latest	Retrieve the latest available financial record for one company.
/companies/{company}/ratios	Review historical ratio data for a company.
/companies/{company}/profitability	Review profitability intelligence.
/companies/{company}/leverage	Review debt, leverage and interest-coverage intelligence.
/companies/{company}/growth	Review revenue, PAT and EPS growth information.
/companies/{company}/cashflow	Review cash-flow intelligence.

Endpoint	Use
/companies/{company}/capital-allocation	Review capital-allocation patterns.
/companies/{company}/cluster	View the company's financial cluster.
/screen	Apply screening thresholds such as minimum ROE and maximum D/E.
/sectors	List sectors and company counts.
/sectors/{sector}/companies	List companies belonging to a sector.
/peers/{company}	Retrieve peer information for a company.

4. How to analyze a company

Step 1 — Identify the company: Use `/companies` and confirm the `ticker/company_id` is available.

Step 2 — Check the latest record: Use `/companies/{company}/latest` to understand the latest sales, profit, assets, borrowings and other available fields.

Step 3 — Review profitability: Check ROE, ROCE, ROA, net profit margin and operating profit margin.

Step 4 — Review leverage: Check debt-to-equity, interest coverage, net debt and the high-leverage flag. If interest expense is zero, the project rule is to label the situation Debt Free rather than divide by zero.

Step 5 — Review growth: Review revenue CAGR, PAT CAGR and EPS CAGR together. If the CAGR base year is negative, the project rule is TURNAROUND instead of calculating a normal CAGR.

Step 6 — Review cash flow: Check free cash flow, CFO quality, CapEx intensity, FCF conversion and the capital-allocation pattern.

Step 7 — Check the cluster: Use `/companies/{company}/cluster` to understand the assigned financial-profile group.

Step 8 — Compare peers: Use `/peers/{company}` and the peer-analysis outputs to compare the company with its relevant peers.

5. Key KPI interpretation

KPI	Interpretation
Net Profit Margin	Net profit as a percentage of sales; higher generally indicates stronger profit conversion.
Operating Profit Margin	Operating profit relative to sales; useful for operating efficiency.
ROE	Return generated on shareholders' equity.
ROCE	Return generated on capital employed.
ROA	Profitability relative to the asset base.
Debt-to-Equity	Borrowings relative to equity; use with sector context.
Interest Coverage	Ability of operating profit to cover interest expense.
Asset Turnover	Sales generated relative to assets.
Free Cash Flow	Cash flow available after the project's defined investing/capital-spending treatment.
FCF Conversion	Free cash flow relative to profit, useful for checking cash conversion.

6. Cash-flow pattern labels

- **Reinvestor** — operating cash positive, investing cash negative, financing cash negative.
- **Liquidating Assets** — operating cash positive, investing cash positive, financing cash negative.
- **Distress Signal** — operating cash negative, investing cash positive, financing cash positive.
- **Growth Funded by Debt** — operating cash negative, investing cash negative, financing cash positive.
- **Mixed** — mixed cash-flow sign combinations covered by the project's rules.
- **Pre-Revenue** — all three cash-flow components negative under the project's classification.

7. Clustering

The current D-19 output contains one row per company in `output/cluster_labels.csv`. The current run produced 93 unique companies and no missing cluster labels. The labels generated in the current run are:

Cluster label	Current run count
Growth & Quality	3
Lower Financial Quality	68
Stable / Moderate	21
Strong Financial Profile	1

8. Screening guidance

Use the screening endpoint to narrow the company universe using financial thresholds. For example, the project exposes parameters such as minimum ROE and maximum debt-to-equity. Interpret results in sector context and remember the project rule that the Financials sector is skipped when applying the D/E screener filter.

Example request:

```
http://127.0.0.1:8000/screen?min_roe=20&max_de=2
```

9. Sector and peer analysis

- Use `/sectors` to see the sector universe and company counts.
- Use `/sectors/{sector}/companies` to identify the companies in a selected sector.
- Use `/peers/{company}` when a company-level comparison is required.
- The project also contains peer-comparison outputs for analyst review.

10. Dashboard workflow

- Start the dashboard with **make dashboard**.
- Use the dashboard to explore company, sector, screening, peer and financial intelligence views.
- Treat simulated market-related datasets such as **stock_prices** and **market_cap** as **SIMULATED**. They must be clearly labelled as such when displayed or discussed.

11. Data and project rules

- Core Excel files must be read with `pd.read_excel(path, header=1)`.
- Normalize **company_id** by trimming spaces and converting to uppercase before joins.
- Monetary values are stored in **INR Crore**.
- Skip the **Financials** sector when applying the D/E screener filter.
- If CAGR has a negative base year, return **TURNAROUND** rather than calculating CAGR.
- If Interest Expense = 0, display **Debt Free** instead of dividing by zero.
- Clearly label simulated **stock_prices** and **market_cap** data as **SIMULATED**.
- Run **make test** before every Git commit. Zero test failures are mandatory.

12. Analyst checklist

Check	Completed?
Confirm API health and database availability	■
Confirm company ID and latest financial record	■
Review profitability KPIs	■
Review leverage and interest coverage	■
Review revenue/PAT/EPS growth	■
Review cash-flow intelligence and capital allocation	■
Review company cluster	■
Compare with peers and sector	■

Check	Completed?
Run the required tests before commit	■

Document status: D-22 Analyst Guide. This guide reflects the project's current API, database, clustering and financial-intelligence workflow as established during project completion.